

You must learn the following definitions and formulae:

- The **sample mean** is $\bar{x} = \frac{\sum x_i}{n}$
- The **sum of the squared deviations from the sample mean** is $S_{xx} = \sum (x_i - \bar{x})^2$
- The **(sample) variance** is $s^2 = \frac{S_{xx}}{n}$
- The **(sample) standard deviation** is $s = \sqrt{\frac{S_{xx}}{n}}$

1. A sample of 11 seed pods has the following number of seeds per pod:

4 5 5 6 8 8 12 14 15 19 25

Calculate the mean and the standard deviation.

2. Two maths classes sit the same exam. Their scores are below.

Class A	47	51	52	53	55	55	57	58	59	61	63	71
Class B	31	37	42	48	49	55	59	62	70	71	72	78

- Find the mean and standard deviation of each set of scores. Write two comments comparing the distribution of marks in the two classes.
- One class was set based on ability, while the other was a random set. Giving a reason, suggest which is which.

You also need to know the alternative formula for S_{xx}

$$S_{xx} = \sum (x_i - \bar{x})^2 = \sum x_i^2 - n\bar{x}^2$$

and how to find the mean, variance, and standard deviation using your calculator's statistical functions.

- Verify your answers to Question 1 overleaf by
 - using your calculator's statistical functions to find the sample mean
 - then using the alternative formula for S_{xx} to find the standard deviation
 - finally, check your answer by using your calculator to find the standard deviation directly.
- A set of thirty numbers are summarised by $\sum x = 350$ and $\sum x^2 = 12375$. Find the mean and variance of the numbers.

5. Students in a hostel were asked how many baths they had taken during a given fortnight. Find
- the mean
 - S_{xx}
 - the standard deviation.

Number of baths	0	1	2	3	4	5	6	7	8	9	10
Frequency	0	2	3	8	4	4	3	6	3	2	1

Check your answers using your calculator's statistical functions.

6. For the following grouped frequency distribution, calculate approximate values of the mean and the standard deviation.

x	8-12	13-17	18-22	23-27	28-32
f	6	8	12	8	6

7. A travel agency has two high street branches, R and S. Some data about the number of holidays purchased from each shop in a particular week is shown in the table below.

	Number of holidays	Mean cost	Standard deviation
Branch R	32	790.35	56.3
Branch S	24	651.85	27.1

Find the mean cost and standard deviation of all 56 holidays purchased.

8. Parplan Opinion Polls Ltd. conducted a nationwide survey into the attitude of teenage girls. One of the questions asked was “What is the ideal age for a girl to have her first baby?”. The respondents were divided into two groups, the Northern zone and the Southern zone.

In reply, the sample of 165 girls from the Northern zone gave a mean of 23.4 years and a standard deviation of 1.6 years.

The overall combined sample of 384 girls in total gave a mean of 24.8 years and a standard deviation of 2.2 years.

What were the mean and standard deviation for the girls from the Southern zone?